

MTH 202 Chapter 1: Three-Dimensional Space & Vectors

1.2 Worksheet

1. Use the given diagram to write each combination of vectors as a single vector.

a) $\overrightarrow{AB} + \overrightarrow{BC}$

$$\overrightarrow{AB} + \overrightarrow{BC} = \overrightarrow{AC}$$

b) $\overrightarrow{CD} + \overrightarrow{DB}$

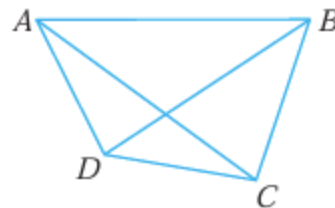
$$\overrightarrow{CD} + \overrightarrow{DB} = \overrightarrow{CB}$$

c) $\overrightarrow{DB} - \overrightarrow{AB}$

$$\overrightarrow{DB} - \overrightarrow{AB} = \overrightarrow{DA}$$

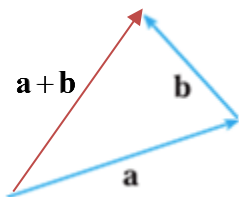
d) $\overrightarrow{DC} + \overrightarrow{CA} + \overrightarrow{AB}$

$$\overrightarrow{DC} + \overrightarrow{CA} + \overrightarrow{AB} = \overrightarrow{DB}$$

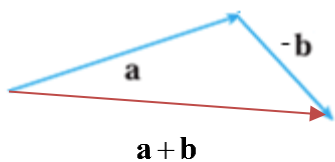


2. Copy the vectors in the figure and use them to draw the following vectors.

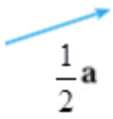
a) $\mathbf{a} + \mathbf{b}$



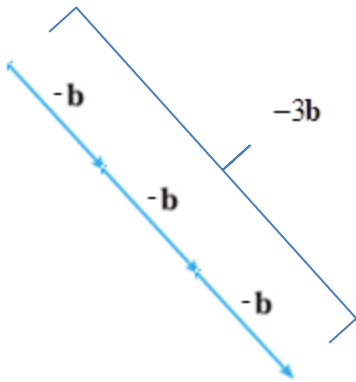
b) $\mathbf{a} - \mathbf{b}$



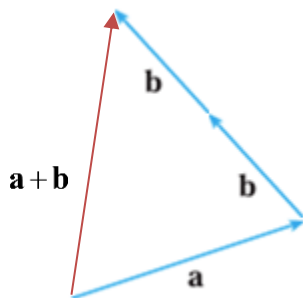
c) $\frac{1}{2}\mathbf{a}$



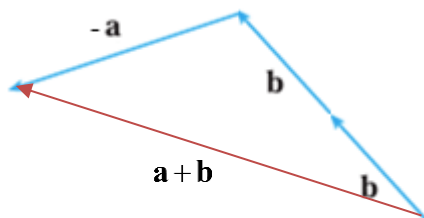
d) $-3\mathbf{b}$



e) $\mathbf{a} + 2\mathbf{b}$



f) $2\mathbf{b} - \mathbf{a}$

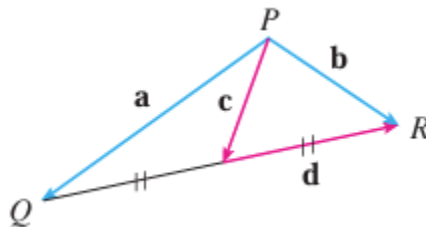


3. In the figure, the tip of $\mathbf{c}$ and the tail of $\mathbf{d}$ are both the midpoint of QR . Express $\mathbf{c}$ and $\mathbf{d}$ in terms of $\mathbf{a}$ and $\mathbf{b}$.

We see that

$$\mathbf{b} = \mathbf{c} + \mathbf{d}$$

$$\mathbf{a} = \mathbf{c} - \mathbf{d}$$



We can then treat this as a system of equations to find formulas for $\mathbf{c}$ and $\mathbf{d}$.

By adding both equations we obtain $\mathbf{a} + \mathbf{b} = 2\mathbf{c} \rightarrow \frac{1}{2}(\mathbf{a} + \mathbf{b}) = \mathbf{c}$.

By subtracting the equations we obtain $\mathbf{b} - \mathbf{a} = 2\mathbf{d} \rightarrow \frac{1}{2}(\mathbf{b} - \mathbf{a}) = \mathbf{d}$.

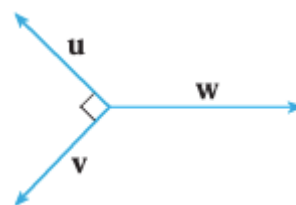
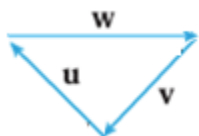
4. If the vectors in the figure satisfy $\|\mathbf{u}\| = \|\mathbf{v}\| = 1$ and $\mathbf{u} + \mathbf{v} + \mathbf{w} = \mathbf{0}$, what is $\|\mathbf{w}\|$?

By redrawing the vectors to illustrate $\mathbf{u} + \mathbf{v} + \mathbf{w} = \mathbf{0}$ we see that the vectors

form a right triangle. Since $\|\mathbf{u}\| = \|\mathbf{v}\| = 1$ we then

have, by the Pythagorean Theorem, that

$$\|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 = \|\mathbf{w}\|^2 \rightarrow 1 + 1 = \|\mathbf{w}\|^2 \rightarrow \|\mathbf{w}\| = \sqrt{2}$$



5. Find $\mathbf{a} + \mathbf{b}$, $2\mathbf{a} + 3\mathbf{b}$, $\|\mathbf{a}\|$, and $\|\mathbf{a} - \mathbf{b}\|$ for each of the following sets of vectors $\mathbf{a}$ and $\mathbf{b}$.

a) $\mathbf{a} = \langle 5, -12 \rangle$ & $\mathbf{b} = \langle -3, -6 \rangle$

$$\mathbf{a} + \mathbf{b} = \langle 5 + (-3), -12 + (-6) \rangle = \langle 2, -18 \rangle$$

$$2\mathbf{a} + 3\mathbf{b} = 2\langle 5, -12 \rangle + 3\langle -3, -6 \rangle = \langle 10, -24 \rangle + \langle -9, -18 \rangle = \langle 1, -42 \rangle$$

$$\|\mathbf{a}\| = \sqrt{(5)^2 + (-12)^2} = \sqrt{25 + 144} = \sqrt{169} = 13$$

$$\|\mathbf{a} - \mathbf{b}\| = \|\langle 5 - (-3), -12 - (-6) \rangle\| = \|\langle 8, -6 \rangle\| = \sqrt{8^2 + (-6)^2} = \sqrt{64 + 36} = \sqrt{100} = 10$$

b) $\mathbf{a} = 4\mathbf{i} + \mathbf{j}$ & $\mathbf{b} = \mathbf{i} - 2\mathbf{j}$

$$\mathbf{a} + \mathbf{b} = 5\mathbf{i} - \mathbf{j} = \langle 5, -1 \rangle$$

$$2\mathbf{a} + 3\mathbf{b} = 2(4\mathbf{i} + \mathbf{j}) + 3(\mathbf{i} - 2\mathbf{j}) = 11\mathbf{i} - 4\mathbf{j} = \langle 11, -4 \rangle$$

$$\|\mathbf{a}\| = \sqrt{(4)^2 + (1)^2} = \sqrt{16+1} = \sqrt{17}$$

$$\|\mathbf{a} - \mathbf{b}\| = \|3\mathbf{i} + 3\mathbf{j}\| = \sqrt{(3)^2 + (3)^2} = \sqrt{9+9} = \sqrt{18} = 3\sqrt{2}$$

c) $\mathbf{a} = \langle 1, 2, -3 \rangle$ & $\mathbf{b} = \langle -2, -1, 5 \rangle$

$$\mathbf{a} + \mathbf{b} = \langle 1 + (-2), 2 + (-1), -3 + 5 \rangle = \langle -1, 1, 2 \rangle$$

$$2\mathbf{a} + 3\mathbf{b} = 2\langle 1, 2, -3 \rangle + 3\langle -2, -1, 5 \rangle = \langle 2, 4, -6 \rangle + \langle -6, -3, 15 \rangle = \langle -4, 1, 9 \rangle$$

$$\|\mathbf{a}\| = \sqrt{(1)^2 + (2)^2 + (-3)^2} = \sqrt{1+4+9} = \sqrt{14}$$

$$\|\mathbf{a} - \mathbf{b}\| = \|\langle 1 - (-2), 2 - (-1), -3 - 5 \rangle\| = \|\langle 3, 3, -8 \rangle\| = \sqrt{(3)^2 + (3)^2 + (-8)^2} = \sqrt{9+9+64} = \sqrt{82}$$

d) $\mathbf{a} = 2\mathbf{i} - 4\mathbf{j} + 4\mathbf{k}$ & $\mathbf{b} = 2\mathbf{j} - \mathbf{k}$

$$\mathbf{a} + \mathbf{b} = 2\mathbf{i} - 2\mathbf{j} + 3\mathbf{k} = \langle 2, -2, 3 \rangle$$

$$2\mathbf{a} + 3\mathbf{b} = 2(2\mathbf{i} - 4\mathbf{j} + 4\mathbf{k}) + 3(2\mathbf{j} - \mathbf{k}) = 4\mathbf{i} - 2\mathbf{j} + \mathbf{k} = \langle 4, -2, 1 \rangle$$

$$\|\mathbf{a}\| = \sqrt{(2)^2 + (-4)^2 + (4)^2} = \sqrt{4+16+16} = \sqrt{36} = 6$$

$$\|\mathbf{a} - \mathbf{b}\| = \|2\mathbf{i} - 6\mathbf{j} + 5\mathbf{k}\| = \sqrt{(2)^2 + (-6)^2 + (5)^2} = \sqrt{4+36+25} = \sqrt{65}$$

6. Find a unit vector that has the same direction as the given vector.

a) $\langle -3, 7 \rangle$

We see that the vector $\langle -3, 7 \rangle$ has a length of $\|\langle -3, 7 \rangle\| = \sqrt{(-3)^2 + (7)^2} = \sqrt{9 + 49} = \sqrt{58}$.

Therefore, the unit vector with the same direction would be $\mathbf{u} = \frac{1}{\sqrt{58}} \langle -3, 7 \rangle = \left\langle \frac{-3}{\sqrt{58}}, \frac{7}{\sqrt{58}} \right\rangle$.

b) $-4\mathbf{i} + 2\mathbf{j} + 4\mathbf{k}$

We see that the vector $-4\mathbf{i} + 2\mathbf{j} + 4\mathbf{k}$ has a length of

$$\|-4\mathbf{i} + 2\mathbf{j} + 4\mathbf{k}\| = \sqrt{(-4)^2 + (2)^2 + (4)^2} = \sqrt{16 + 4 + 16} = \sqrt{36} = 6.$$

Therefore, the unit vector with the same direction would be

$$\mathbf{u} = \frac{1}{6}(-4\mathbf{i} + 2\mathbf{j} + 4\mathbf{k}) = -\frac{2}{3}\mathbf{i} + \frac{1}{3}\mathbf{j} + \frac{2}{3}\mathbf{k} = \left\langle -\frac{2}{3}, \frac{1}{3}, \frac{2}{3} \right\rangle.$$

7. Find a vector that has the same direction as $\langle -2, 4, 2 \rangle$ but has a length of 6.

We see that the vector $\langle -2, 4, 2 \rangle$ has a length of

$$\|\langle -2, 4, 2 \rangle\| = \sqrt{(-2)^2 + (4)^2 + (2)^2} = \sqrt{4 + 16 + 4} = \sqrt{24} = 2\sqrt{6}.$$

Therefore, the unit vector with the same direction would be $\mathbf{u} = \frac{1}{2\sqrt{6}} \langle -2, 4, 2 \rangle = \left\langle -\frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}}, \frac{1}{\sqrt{6}} \right\rangle$.

Thus, if we scale this vector to now have a length of 6 we get

$$\mathbf{v} = 6 \left\langle -\frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}}, \frac{1}{\sqrt{6}} \right\rangle = \left\langle -\frac{6}{\sqrt{6}}, \frac{12}{\sqrt{6}}, \frac{6}{\sqrt{6}} \right\rangle = \langle -\sqrt{6}, 2\sqrt{6}, \sqrt{6} \rangle$$